KINDNS

KINDNS:

A framework to improve the security and resilience of the DNS

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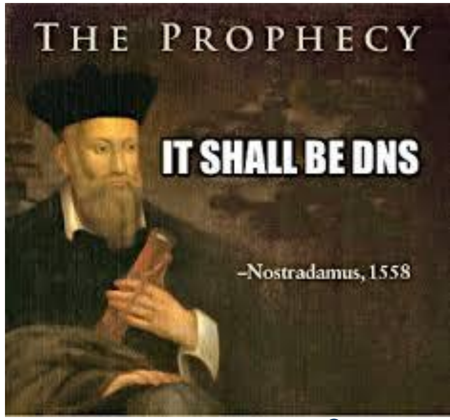
It's always DNS!

- *a popular meme among network professionals that refers to how often the Domain Name System (DNS) is the root cause of network problems*
- *DNS: a foundational technology*
  - *critical to almost all internet and network functions*
  - *Simple (in theory ? – resolves names into numbers), but (incredibly) complex to manage, welcome misconfigurations and failures.*
  - *See https://powerdns.org/dns-camel/ and https://ianix.com/pub/dnssec-outages.html for a good headache!*



The DNS glues the Internet together

Operators scratching their heads

… DNS software developers too

… and any of us trying to raise awareness and help operators to be good citizens and safely ride the Camel too.

… the idea of KINDNS came from that observation

How kind is your DNS to the Internet ?

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- *DNS translates human-readable names into IP addresses.*
- *A **critical function** for the Internet and its services (emails, web, cloud services, online payments and services, etc.)*
- ***Contributes to the security and performance of the Internet**: distributed functions, caching, DNSSEC, other security mechanisms.*
- ***DNS incidents** can impact organizations: reputation, productivity, service disruption, data breaches, etc.*
    - *DNS as attack vector, DNS misconfiguration or vulnerabilities.*
    - *Low/moderate/severe/critical/local/large, difficult to measure impacts.*

What is KINDNS ?

Knowledge-sharing and **I**nstantiating **N**orms for **D**NS (Domain Name System) and **N**aming **S**ecurity

*A simple **framework** that can **help a wide variety of DNS operators**, from small to large, to follow both the **evolution of the DNS** protocol and the best practices that the industry identifies for better security and more effective DNS operations.*

*….. is pronounced "**kindness**"*

The Four Components

THE OPERATORS CATEGORIES

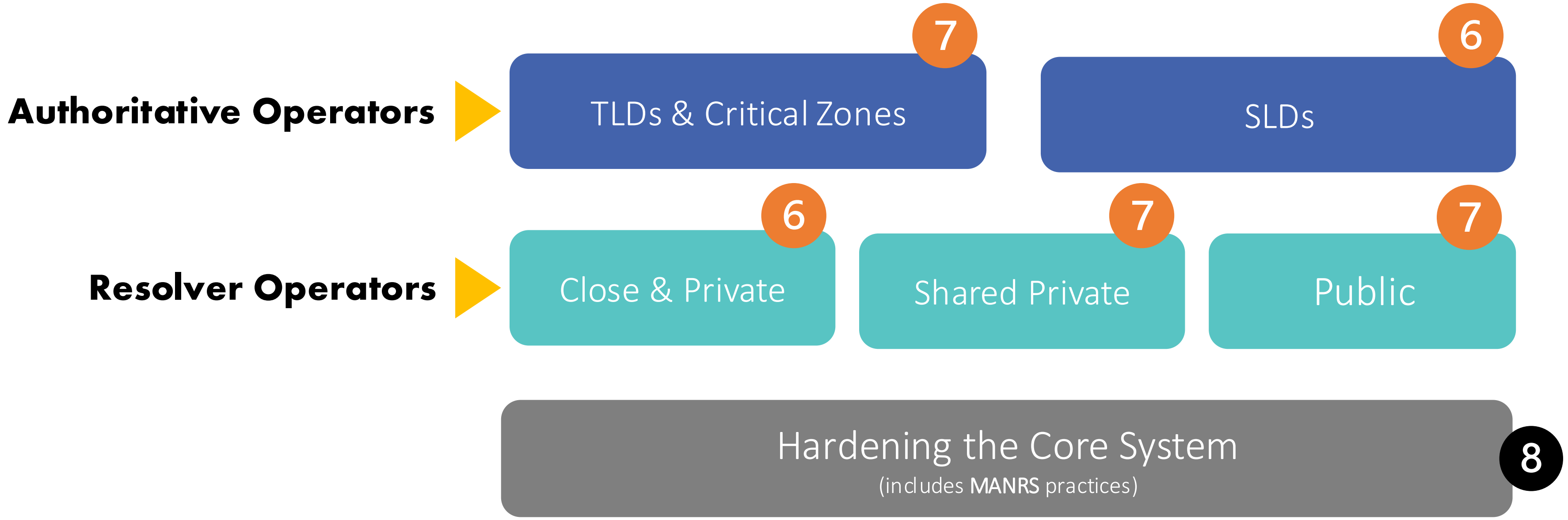
THE GUIDEBOOK

THE SELF-ASSESSMENT TOOL & ITS REPORT

THE ENROLLMENT AS SUPPORTER

compliance and commitment to share the good will

Targeted Operators



- Each category has **6-8 practices** that we encourage operators to implement & **a detailed guidebook** on why and how to implement them. *See www.kindns.org, for more details*
- By joining KINDNS, DNS operators are voluntarily committing to adhere to these identified practices and act as "goodwill ambassadors" within the community.

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Practices for Authoritative DNS Operators of Critical Zones

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TLDs & Critical Zones

1. **MUST** be DNS Security Extensions (DNSSEC) signed and follow key management best practices.
2. Transfer between authoritative servers **MUST** be limited
3. Zone file integrity **MUST** be controlled
4. Authoritative and recursive nameservers **MUST run on separate infrastructure**
5. A minimum of two distinct nameservers **MUST** be used for any given zone
6. There **MUST** be diversity in the operational infrastructure: **Network, Geographical, Software**
7. The infrastructure that makes up your DNS infrastructure **MUST** be monitored

Practices for Authoritative DNS Operators of SLDs

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SLDs

1. **MUST** be DNSSEC signed and follow key management best practices
2. Transfer between authoritative servers **MUST** be limited
3. Zone file integrity **MUST** be controlled
4. Authoritative and recursive nameservers **MUST run on separate infrastructure**
5. A minimum of two distinct nameservers **MUST** be used for any given zone
6. Authoritative servers for a given zone **MUST** run from diversified infrastructure
7. The infrastructure that makes up your DNS infrastructure **MUST** be monitored

Practices for Closed & Private Resolver Operators

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*Private resolvers are **not publicly accessible** and cannot be reached over the open internet. They are typically found in corporate networks or other restricted-access networks*

Closed & Private resolvers

1. DNSSEC validation **MUST** be enabled
2. Access control list (ACL) statements **MUST** be used to restrict who may send recursive queries
3. QNAME minimization **MUST** be enabled
4. Authoritative and recursive nameservers **MUST** run on separate infrastructure
5. At least two distinct servers **MUST** be used for providing recursion services
6. Authoritative servers for a given zone **MUST** run from a diversified Infrastructure
7. The infrastructure that makes up your DNS infrastructure **MUST** be monitored

Practices for Shared Private Resolver Operators

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Shared private resolver operators are typically ISPs or similar hosting service providers. They offer DNS resolution services to their customers (mobile, cable/DSL/fiber users, as well as hosted servers and applications).

Shared Private resolvers

1. DNSSEC validation **MUST** be enabled
2. ACL statements **MUST** be used to restrict who may send recursive queries
3. QNAME minimization **MUST** be enabled
4. Authoritative and recursive nameservers **MUST** run on separate infrastructure
5. At least two distinct servers **MUST** be used for providing recursion services
6. The infrastructure that make up your DNS infrastructure **MUST** be monitored
7. For privacy consideration: Encryption (DOH or DoT) **SHOULD** be enabled
8. Private resolver operators **SHOULD** have software diversity

Practices for Hardening the Core

In addition to implementing best practices for DNS security and for DNS availability and resilience, all operators must pay ***careful attention to practices for hardening the platforms*** *their DNS services use.*

Core Hardening

1. ACLs **MUST** be implemented to control network traffic to your DNS servers

2. BCP38/MANRS egress filtering **MUST** be implemented

3. The configuration of each DNS server **MUST** be locked down

4. User permissions and application access to system resources **MUST** be limited

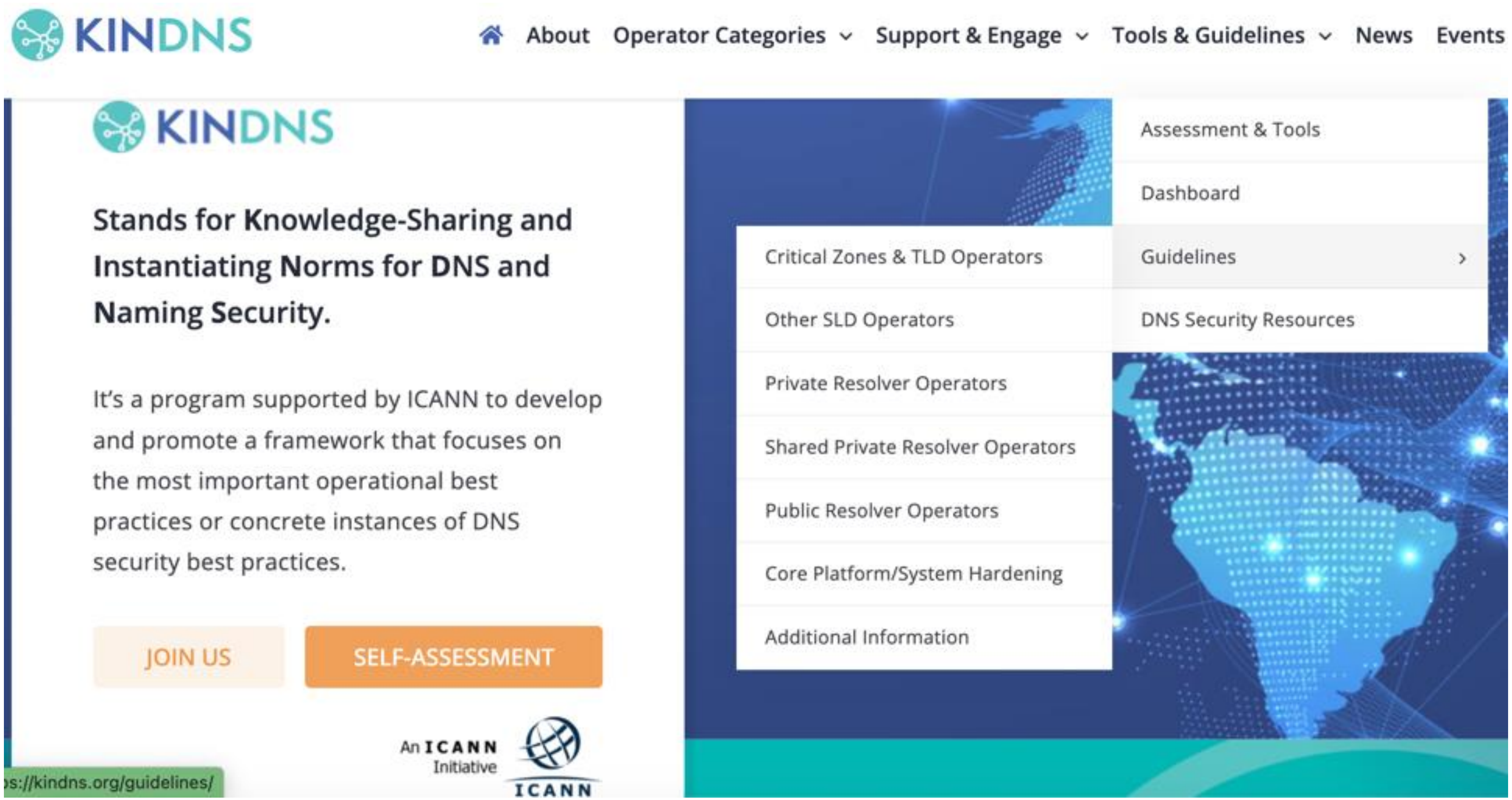
5. System and service configuration files **MUST** be versioned

6. Access to management services **MUST** be restricted

7. Access to the system console **MUST** be secured using cryptographic keys and/or two factor authentication mechanism.

8. Credentials Management for customer access **MUST** adhere to best practices

Practices are all documented



Self-assessment & Enrollment

1. Operators in each category can self-assess their operational practices against KINDNS and use the report to correct/adjust unaligned practices.

    - self-assessment is **anonymous**
    - **reports** can be downloaded directly from the web site.

2. Operators can enroll as participant to one or many categories covered by KINDNS.

    - Participation in the KINDNS initiative means voluntarily committing to implement/adhere to agreed practices.
    - Participants become goodwill ambassadors and promote best practices.

Participation/Enrollment

KINDNS | About | Operator Categories | Support & Engage | Tools & Guidelines | News | Events | JOIN US

Participants

Home » Participants

Operators who voluntarily agreed to implement the KINDNS framework can formally join the initiative by filling out the enrollment form. As a member, you are showing your commitment to implementing the practices promoted by KINDNS in your DNS operation and your support for a robust DNS operation generally.

The following organizations and operators have already joined the initiative, either as supporters or as implementers in the different categories.

KINDNS Participants

TLD & Critical Zone Operators | Other SLD Operators | Private Resolver Operators | Shared Private Resolver Operators | Public Resolver Operators

Organization Name	Practice-1	Practice-2	Practice-3	Practice-4	Practice-5	Practice-6	Practice-7
.PT	✓	✓	✓	✓	✓	✓	✓
Posix (for edu.za)	✓	✓	✓	✓	✓	✓	✓
APNIC	✓	✓	✓	✓	✓	✓	✓
DNS Guru	✓	✓	✓	✓	✓	✓	✓
ZDNS	✓	✓	✓	✓	✓	✓	✓
RICTA (.RW)	✓	✓	✓	✓	✓	✓	✓
AFRINIC	✓	✓	✓	✓	✓	✓	✓
RR64 Brazil	✓	✓	✓	✓	✓	✓	✓

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Organization Name	Practice-1	Practice-2	Practice-3	Practice-4	Practice-5	Practice-6
Posix (for posix.co.za)	✓	✓	✓	✓	✓	✓
ICANN (for icann.org)	✓	✓	✓	✓	✓	✓
PowerNet Solutions	✓	✓	✓	✓	✓	✓
Beijing Guoxu Network Technology	✓	✓	✓	✓	✓	✓

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Organization Name	Practice-1	Practice-2	Practice-3	Practice-4	Practice-5	Practice-6	Practice-7
i8 Digital Brazil	✓	✓	✓	✓	✓	✓	✓
RNP Brazil – EduDNS	✓	✓	✓	✓	✓	✓	
Brazil TecPar	✓	✓	✓	✓	✓	✓	✓
PowerNet Solutions	✓	✓	✓	✓	✓	✓	✓
Intercol Brazil	✓	✓	✓	✓	✓	✓	✓
Tempus Group, Paraguay	✓	✓	✓	✓	✓	✓	✓
NETServ	✓	✓	✓	✓	✓	✓	✓
REDESIM	✓	✓	✓	✓	✓	✓	✓
MTN Ghana	✓	✓	✓	✓	✓	✓	✓
LCI Telecom	✓	✓	✓	✓	✓	✓	✓
RR64 Brazil	✓	✓	✓	✓	✓	✓	✓
RUPI Telecom	✓	✓	✓	✓	✓	✓	✓

2621
of 2656 responses

40%
Completion Rate

12:15 Minutes
Duration

High uptake for Self-assessment

6516	3404	1062	304
Views	**Starts**	**Partial responses**	**Unfinished responses**
Number of times your questionnaire has been opened by someone	Number of times someone clicked the start button of the questionnaire	Number of responses saved with save and continue, but not finished yet	Number of times someone exited questionnaire without a complete response

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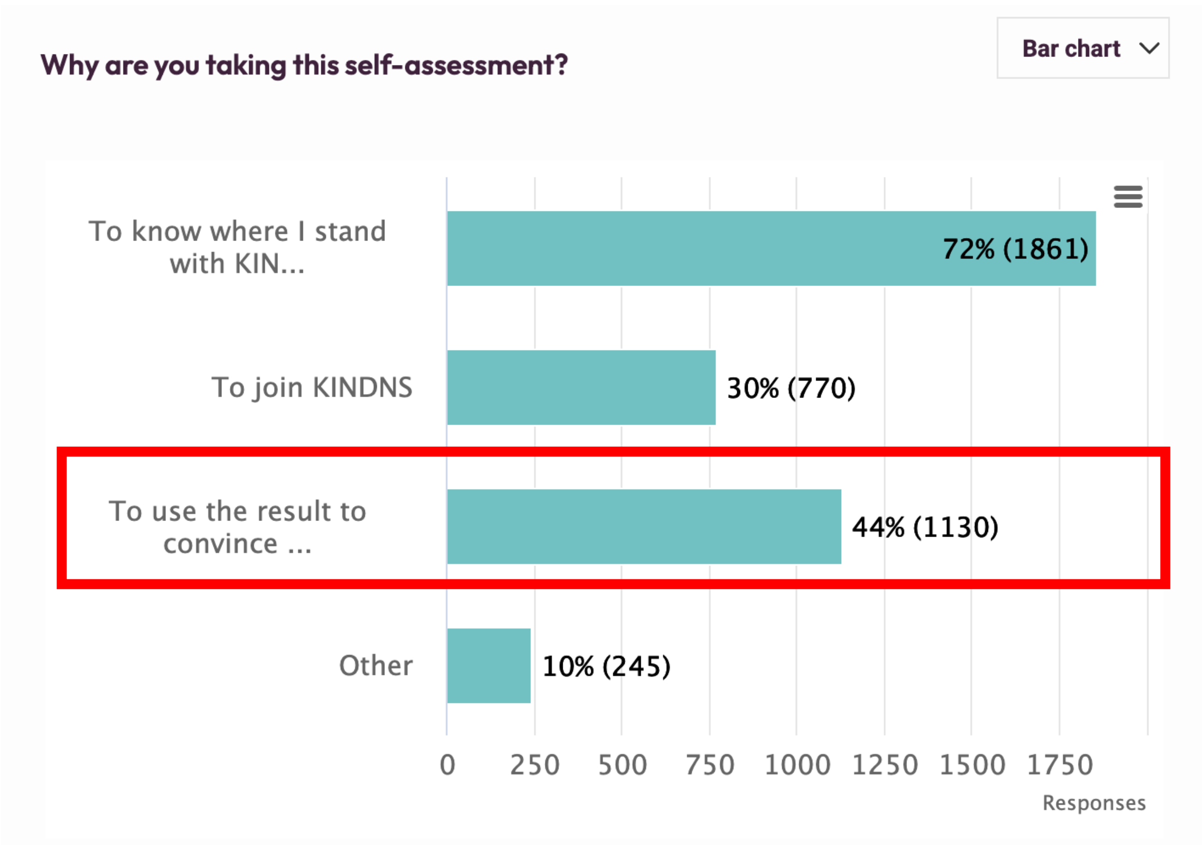
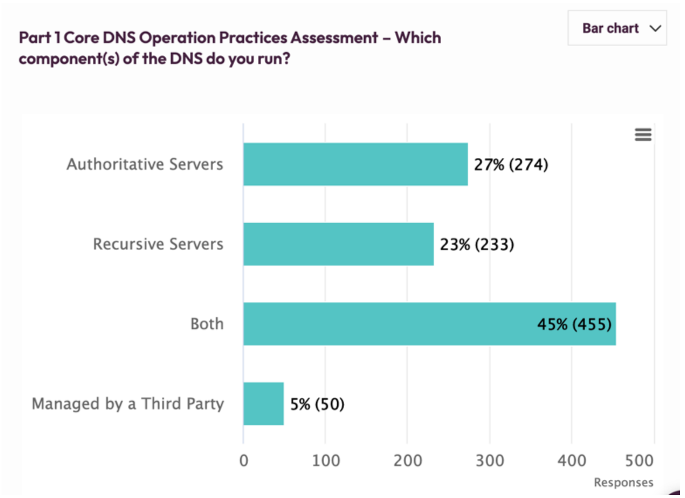
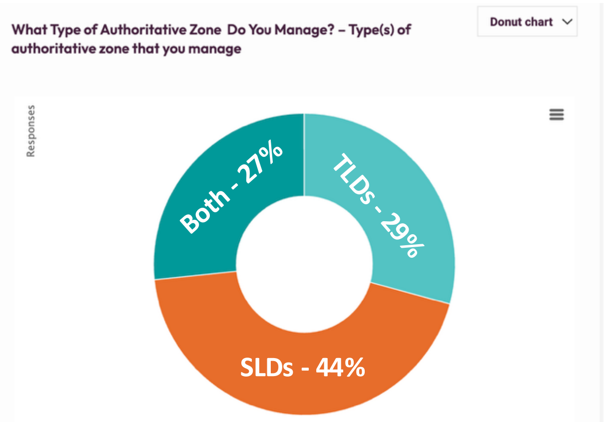

Why are you taking this self-assessment?
Bar chart
To know where I stand with KIN...
72% (1861)
To join KINDNS
30% (770)
To use the result to convince ...
44% (1130)
Other
10% (245)
0
250
500
750
1000
1250
1500
1750
Responses


Part 1 Core DNS Operation Practices Assessment – Which component(s) of the DNS do you run?
Bar chart
Authoritative Servers
27% (274)
Recursive Servers
23% (233)
Both
45% (455)
Managed by a Third Party
5% (50)
0
100
200
300
400
500
Responses


What Type of Authoritative Zone Do You Manage? – Type(s) of authoritative zone that you manage
Donut chart
Responses
Both - 27%
TLDs - 29%
SLDs - 44%

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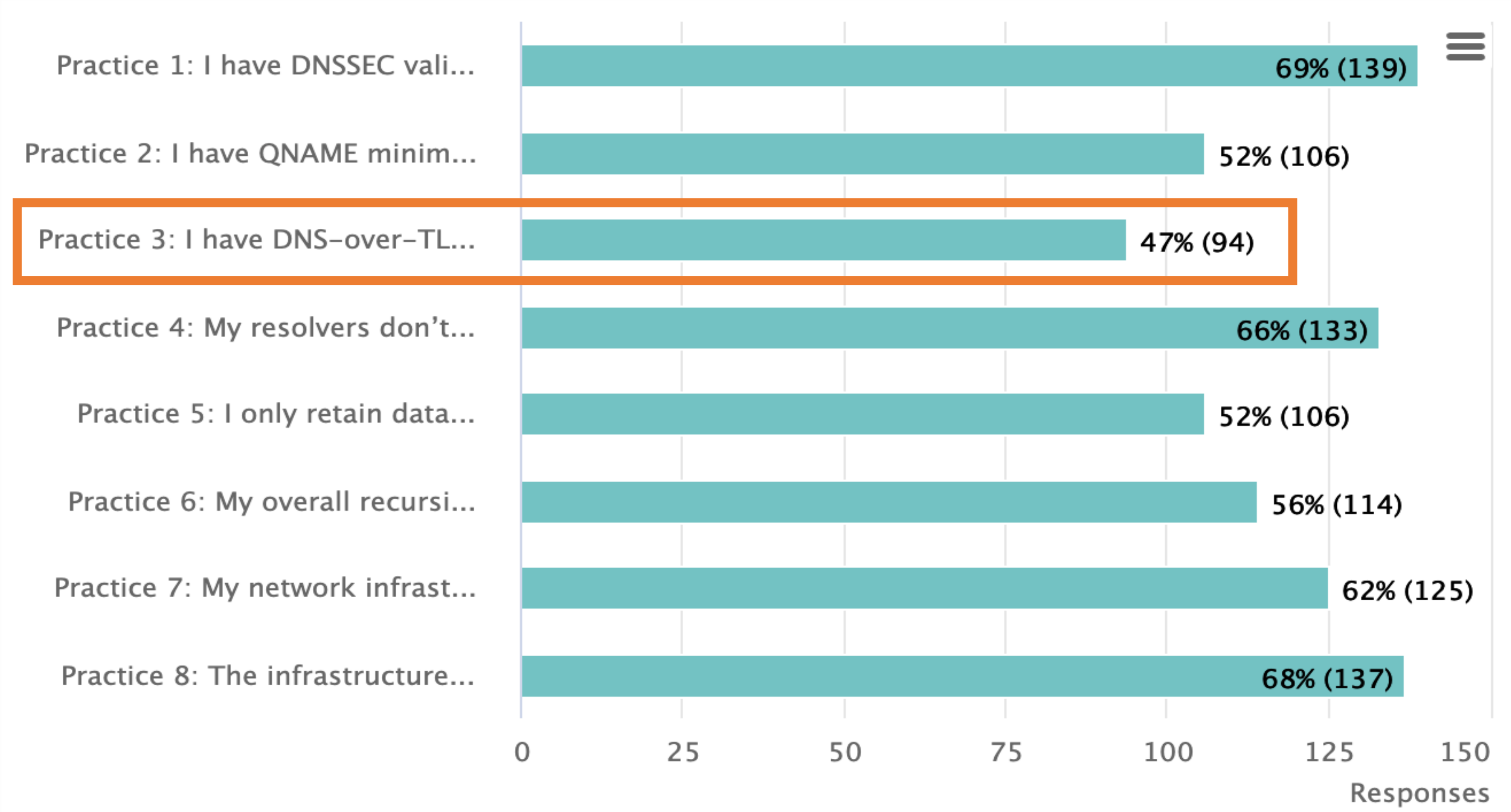
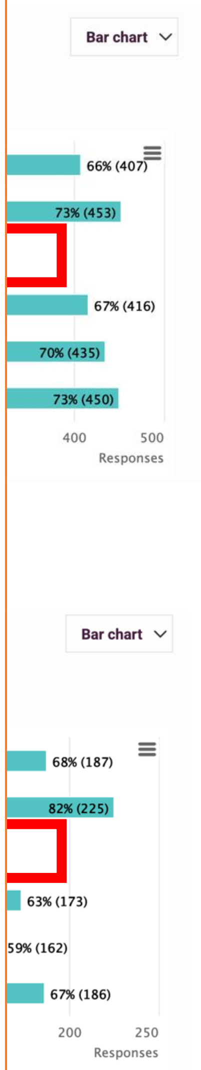
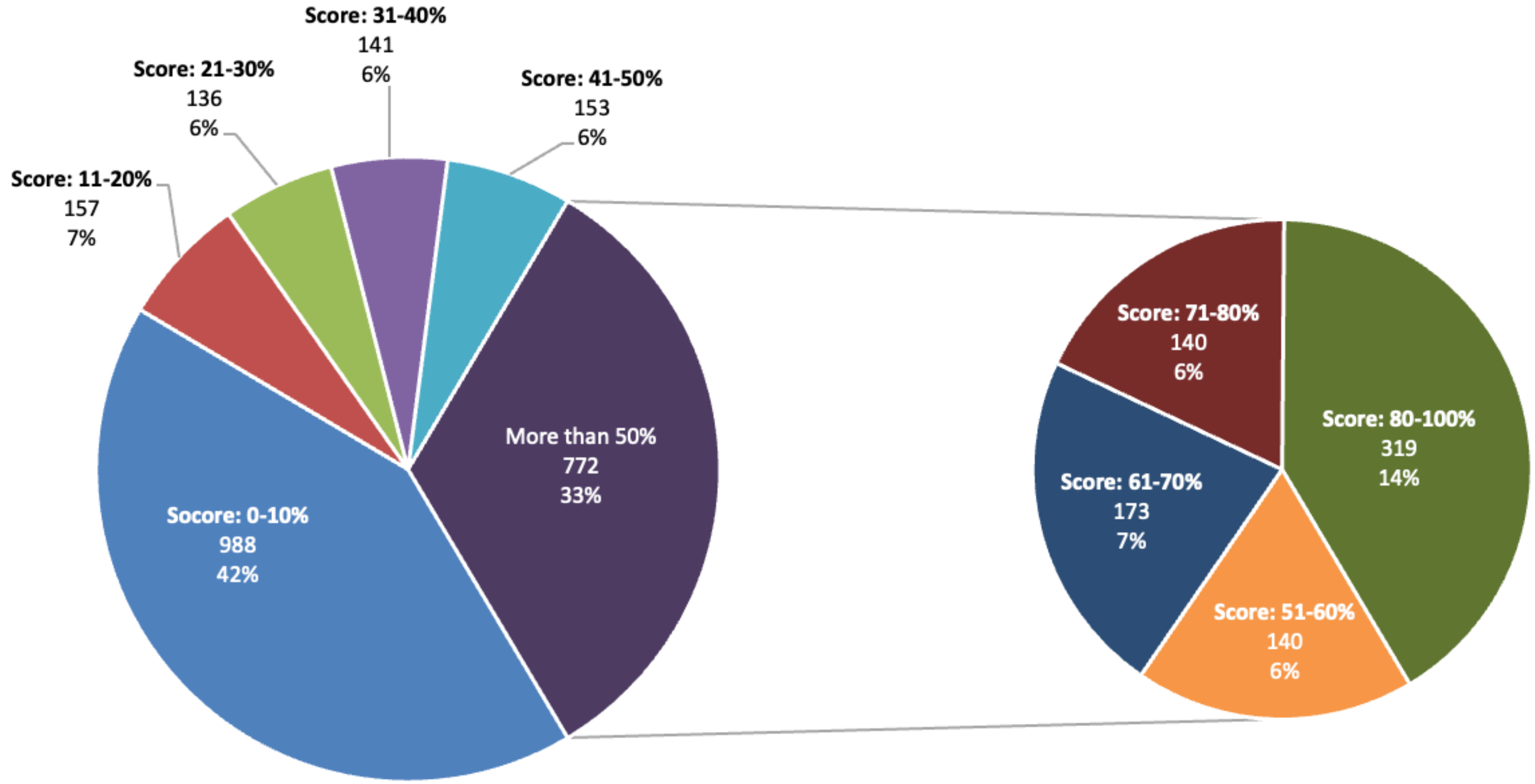
What type
resolvers y

As operator of a Public Recursive Resolver, I implement or adhere to the following practices :





Self-Assessment Ranking



Selecting BCPs

How do we identify them ?

- Draw from own operational experience
- Ask operators (NOG lists, communities)
- Review RFCs and other standards
  https://powerdns.org/dns-camel/
- Shortlist based on relevance, ease of implementation, and how widespread the adoption is

Ask operators to review the selection (kindns-discuss)

Debate and justify choices

Operators must agree on the selected BCPs

kindns-discuss list launched in 2021

- Encouraged operators from all backgrounds to join
- When in doubt, we asked community for advice on what they consider to be a BCP or not
- Some things were debated – is DNSSEC validation a **MUST** nowadays ? (We think so ☺)
- Some practices weren't implemented widely enough, or too complicated (not low hanging fruit) for small operators
  - e.g. Anycast

Current Focus: Phase 2

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Service Platform Hardening

Front-end

- Re-Activate the full enrollment form
- Translate the website and the tools into other languages
- Evolve the Self-assessment to technically measure/assess practices implementation.
  - Two views: Internal & External
  - measure implementation by collecting anonymized data from the self-assessment tool.
  - Integrate a Zonemaster version for Authoritative servers

Back-end

- Integrate the KINDNS server to ICANN E&I monitoring service
- Implement a ticketing system to better track interactions with the public.
- Improve the security fence around WordPress
- Deploy an integrated enrollment management tool (a WP plugin)
- Renew ICANN infosec assessment.
- Directly link self-assessment to enrollment
- Develop an integrated tool to simplify/automate Operator compliance assessment

Current Focus: Phase 2 (con't)

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Community Engagement

Community engagement - continue to encourage operators to get onboard to contribute and support the framework:

- Direct 1:1 Engagements
- Convince/Encourage more DNS operators to join
- Workshops & webinars to raise awareness on KINDNS practices as part of our overall DNS ecosystem security awareness program.
- DNSAthons around secure DNS operations
- Develop partnerships with programs such as MANRS and Pulse, internet.nl, etc.

Communication: a more active communication plan to further promote KINDNS

- Publish a series of DNS best practices dedicated blogs
- Develop toolkits to help operators engage with internal decision-makers.

KINDNS v.2 - Discussion Points

1. **Adding Response Rate Limiting (RRL)** to Authoritative Servers' practice
   - ccTLD and critical Zone Operators
   - Other SLDs too?
2. **Addressing 'Split' responsibilities** for Authoritative servers' operation:
   - Zone file content is controlled by a third party. i.e root server operators and the root zone itself.
3. **Access reliability:** Reachability over IPv6, RPKI for the prefix used for the DNS servers.
4. **Community review team**: volunteers from the community to work with staff to help with assessing participating candidates or other aspect of KINDNS practice evolution.
5. **Metrics**: help measure the impact of KINDNS adoption on global DNS operations

Stay Informed and Contribute

Website	www.kindns.org
Twitter	https://twitter.com/4KINDNS
E-Mail	info@kindns.org
Mailing list	kindns-discuss@icann.org https://mm.icann.org/mailman/listinfo/kindns-discuss

Open discussion

Resilience of your DNS infrastructure

1. Which DNS service do you run ? Authoritative, recursive resolver, both ?

2. How can you check the resilience of the infrastructure ?
   - Diversity: network, location, software.
   - Reachability, availability

3. What about the servers and service performance ?

Protecting your DNS infrastructure ?

1. Are you protecting your DNS infrastructure ?
2. If yes:
   - How are you doing that ?
   - How can that be checked/verified ?

DNSSEC ?

1. Is your domain name DNSSEC signed ? and/or your resolvers do validation ?

2. How to check/confirm that ?

Assess your DNS infrastructure now

Run the KINDNS assessment for your DNS infrastructure:

https://kindns.org/self-assessment/

Engage with ICANN

Thank You and Questions

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